

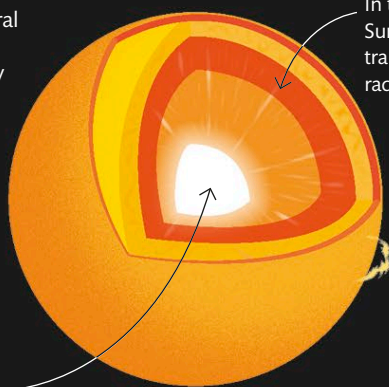
POWERED BY THE SUN

SOLAR ENERGY

The Sun radiates massive amounts of light and heat energy, called solar energy, which is essential for sustaining life on Earth. Solar energy can be converted into useful electrical energy with the help of technologies such as solar cells, which are arranged together to form solar panels. Solar energy is a rapidly growing source of clean, renewable energy, and it is likely to be the world's main energy source by 2050.

THE SUN

The Sun is Earth's closest star and is made of several layers of extremely hot plasma. The Sun's energy is generated by nuclear reactions in its core and then radiates out into space and toward Earth.



In this part of the Sun, energy is transferred by radiation.

In the Sun's core, energy is generated through a process called nuclear fusion.

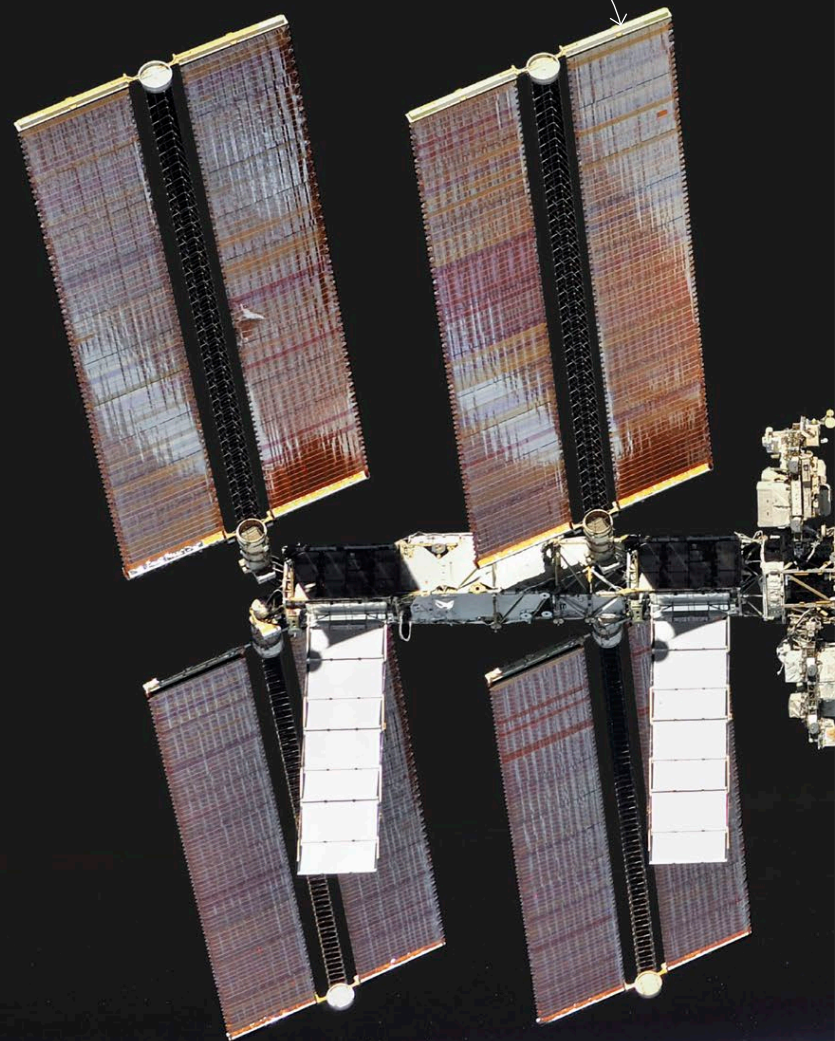
Converting energy from the Sun

Solar cells are arranged in flat solar panels in order to convert the maximum amount of light energy from the Sun into electrical energy. When sunlight falls on a cell, the energy is absorbed, causing electrons to be released from atoms. This produces a flow of electrons through the panel, generating an electric current (see page 34).

SUN



Each solar panel array is 112 ft (34 m) long.



Solar boat

Solar panels can be used to power transportation including the MS *Tûranor PlanetSolar*, pictured here—the world's largest solar-powered boat. It is covered with more than 5,382 sq ft (500 sq m) of solar panels.